Health-Project 3: Breast Cancer Detection

Experiment 4: Data Exploration (EDA)

Course: CSE4001 — Applications of Machine Learning in Industry | **Target:** Malignant (1) vs. Benign (0)

1. Aim

To perform Exploratory Data Analysis (EDA) on the Breast Cancer Wisconsin (Diagnostic) dataset to understand feature distributions, identify relationships between features, detect outliers, and compare the characteristics of benign and malignant tumours.

2. Objectives

1. Analyze the distribution of the 30 numerical features across benign and malignant cohorts.
2. Compare feature values between benign and malignant cases to detect morphometric discriminators.
3. Study correlations between different features and identify multicollinear groups.
4. Detect possible outliers using IQR analysis and boxplot visualizations.
5. Visualize important features using publication-quality charts (histograms, boxplots, heatmaps, scatter plots).
6. Compare features such as radius, texture, perimeter, and area between the two classes.
7. Identify important patterns that can guide model selection in Experiment 5.

3. Dataset Summary

Metric / Property	Value	Description / Clinical Context
Dataset Name	WDBC (Wisconsin)	Digitized Fine-Needle Aspirate (FNA) nuclear images
Sample Count (N)	569 biopsy samples	357 Benign (62.74%) | 212 Malignant (37.26%)
Feature Space	30 continuous	10 nuclear geometries × 3 statistics (mean, se, worst)
Target Variable	Binary (B/0, M/1)	0: Benign (negative) | 1: Malignant (positive)
Class Imbalance	1.68 : 1 ratio	Requires recall & ROC-AUC optimization over raw accuracy

4. Class-Wise Nuclear Morphometric Differences

Nuclear Feature	Benign Mean (B)	Malignant Mean (M)	Relative Increase	Clinical Implication
radius_mean	12.15 µm	17.46 µm	+43.7%	Nuclear enlargement (anisonucleosis)
texture_mean	17.92	21.61	+20.6%	Coarser chromatin clumping
perimeter_mean	78.08 µm	115.37 µm	+47.8%	Irregular nuclear boundary stretch
area_mean	462.79 µm²	978.38 µm²	**+111.4%**	Massive nuclear hypertrophy
concavity_mean	0.046	0.161	**+250.0%**	Severe contour indentations & creases
concave_points_mean	0.026	0.088	**+238.5%**	High frequency of structural folds
compactness_mean	0.080	0.145	+81.3%	Deviations from spherical symmetry

5. Visualizations & Analytical Charts

Figure 1: Morphological Distribution Boxplots: Benign vs. Malignant Tumours

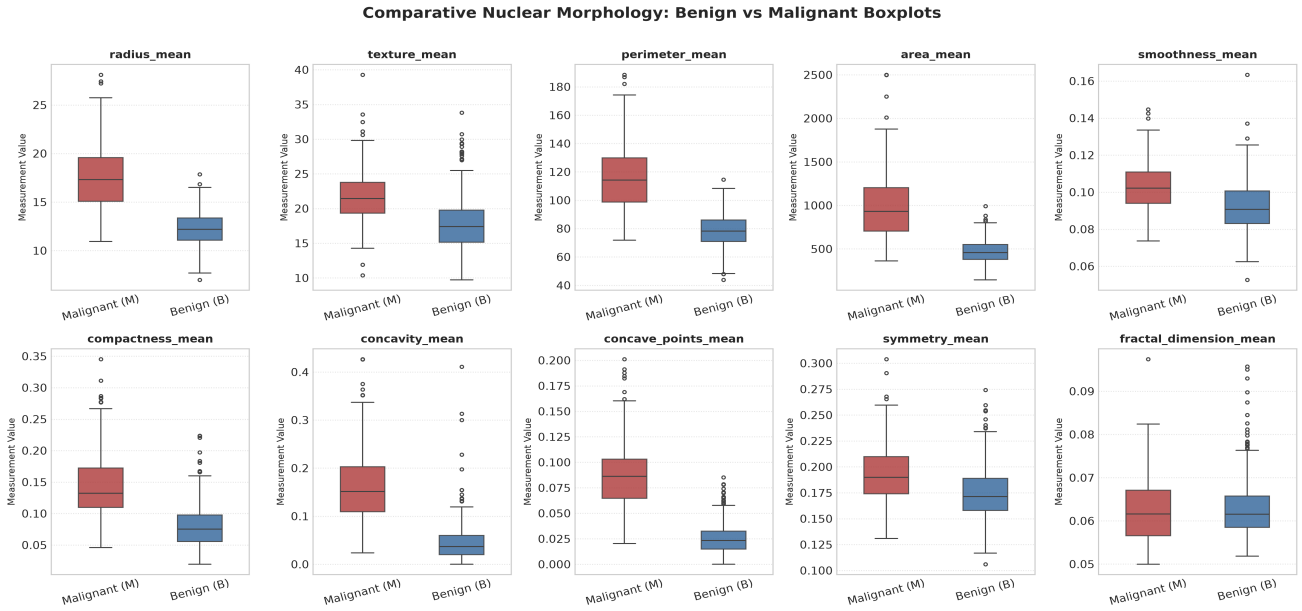
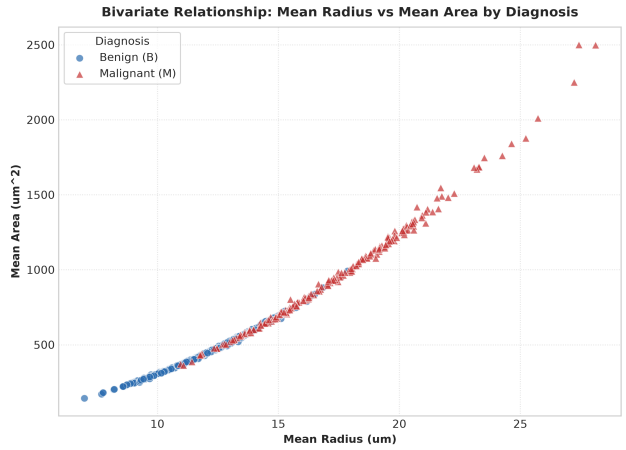


Figure 2: Bivariate Decision Separability (Radius vs. Area) & Class-Wise Averages



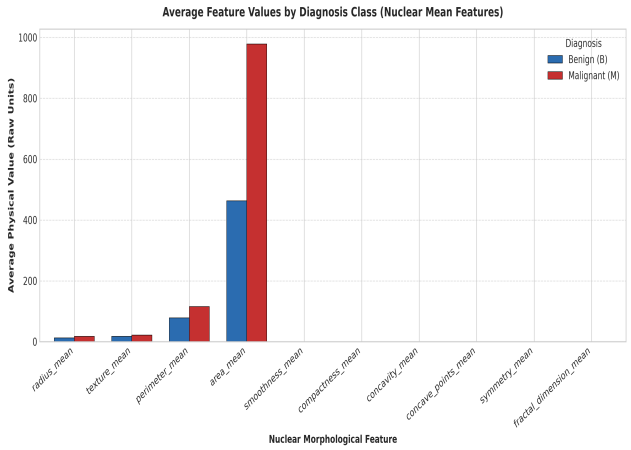


Figure 3: Pearson Correlation Heatmap across 30 Continuous Nuclear Measurements

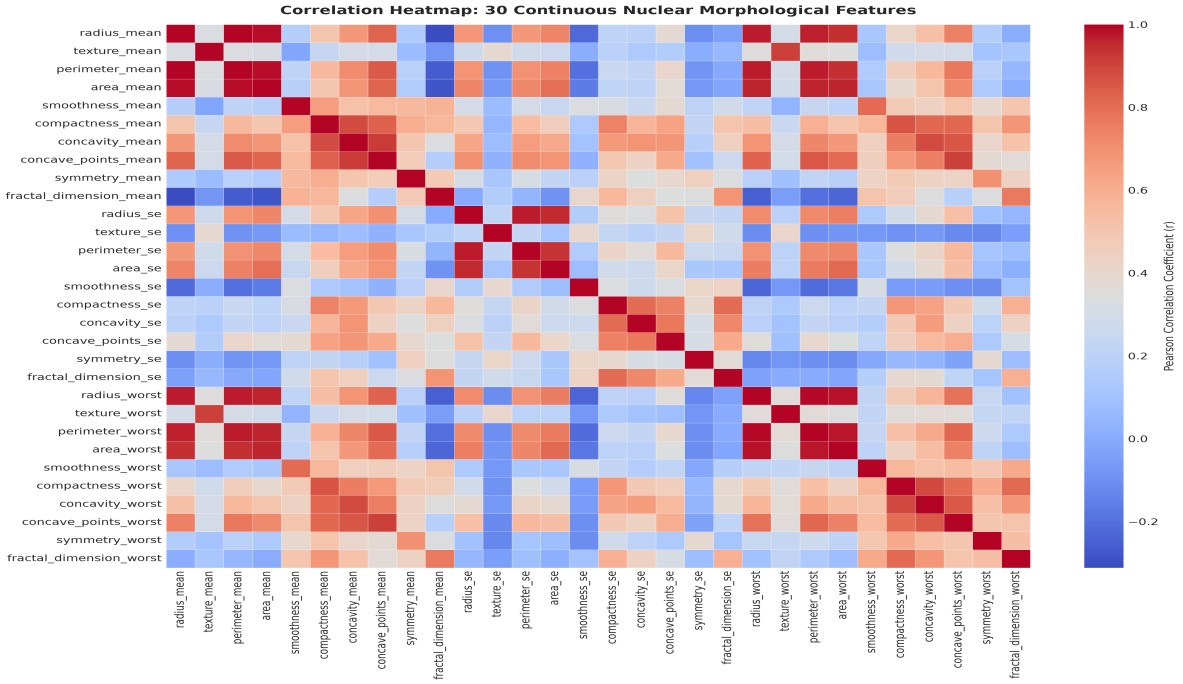
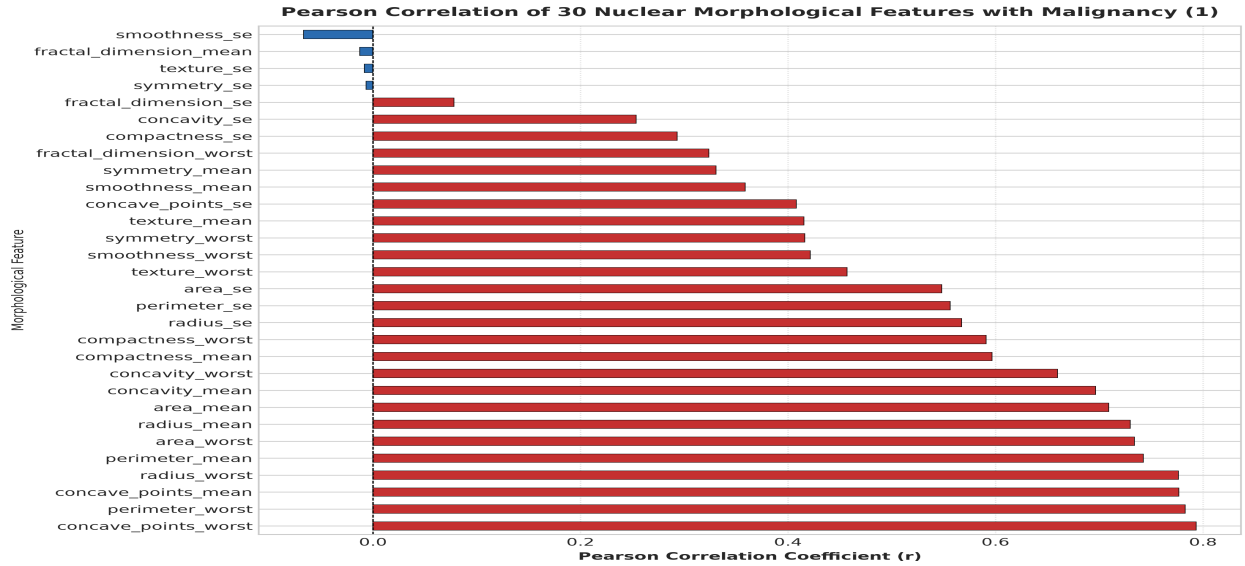


Figure 4: Top Nuclear Morphological Correlates with Malignancy (Target = 1)



6. Multicollinear Feature Redundancy (|r| > 0.90)

Feature Pair (Multicollinear)	Pearson (r)	Redundancy Rationale & Handling Strategy
perimeter_mean ↔ radius_mean	0.9979	Direct geometric dependence (P ≈ 2πr). Handled via L2 regularization.
area_mean ↔ radius_mean	0.9874	Direct geometric dependence (A ≈ πr²). Multicollinear in linear models.
perimeter_worst ↔ radius_worst	0.9937	Collinear in extreme tumor boundary measurements.
area_worst ↔ radius_worst	0.9840	Collinear in worst-case nuclear size metrics.
concave_points_mean ↔ concavity_mean	0.9214	Strong correlation between number and severity of nuclear contour creases.

7. Python EDA Execution Implementation

```
# Correlation Matrix & Multicollinearity Filtering
correlation = df.select_dtypes(include=np.number).corr()
high_corr = [(c1, c2, correlation.loc[c1, c2]) for c1 in correlation.columns
for c2 in correlation.columns if c1 != c2 and abs(correlation.loc[c1, c2]) > 0.90]

# Class-Wise Comparison & Target Correlation
class_means = df.groupby('diagnosis')[features].mean()
target_corr = df.corr(numeric_only=True)['diagnosis'].sort_values(ascending=False)
```

8. Key Observations & Analytical Findings

- **Cohort Demographics:** The dataset contains 569 samples belonging to two diagnosis classes: 357 benign (62.74%) and 212 malignant (37.26%).
- **Class Imbalance Rationale:** The classes are moderately imbalanced (1.68:1 ratio). Downstream evaluation must focus on Malignant Class Recall and ROC-AUC.
- **Pronounced Morphometric Disparity:** Features such as mean radius, perimeter, and area show large differences across classes; malignant nuclei exhibit +111.4% higher mean area.
- **Multicollinearity in Geometry:** Size-related features exhibit extreme correlations (r > 0.98 for radius, perimeter, and area), necessitating regularized loss or ensemble estimators.
- **Contour Indentation as Chief Correlate:** Concave points (worst: r=0.7936, mean: r=0.7766) and concavity exhibit the highest predictive correlations with malignancy.
- **Biological Outliers:** Boxplots reveal extreme observations in high-grade carcinoma instances (area_se in 65 samples); these are preserved as clinically valid signals.
- **Clear Bivariate Separability:** Feature pairings (e.g. Mean Radius vs Mean Area) demonstrate clear separation boundaries between benign and malignant cases.
- **Model Selection Recommendations for Exp 5:** High dimensionality and multicollinearity indicate that Regularized Logistic Regression (L2), Support Vector Machines (RBF Kernel), and Random Forest are ideal baseline candidates.

9. Result

Exploratory Data Analysis was successfully performed on the Breast Cancer Wisconsin dataset. The distributions of important tumour-related features were analyzed, relationships between numerical features were examined using correlation analysis, and differences between benign and malignant cases were visualized using boxplots, scatter plots, and heatmaps.

10. Conclusion

The EDA reveals that several numerical measurements contain useful information for distinguishing between benign and malignant tumour samples. Strong correlations are also present among some features, indicating possible redundancy. These findings directly guide feature selection, regularization, and model training in Experiment 5.